Market Research Document (MRD)

Name of Product: **SignSense**

(Real-time sign language to narrative language translation platform for empowering communication with deaf people)

SignSense
Empowering inclusive communication

Team Number: **10**

Team Members :

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Vision

For **the deaf, hearing-impaired community & people interacting with them**,
who **face communication barriers in their daily interactions**,
our product '**SignSense' is an innovative communication platform**
that provides **real-time translation of sign language into narrative/spoken language and vice versa**.
Unlike existing solutions,
We offer **a user-friendly platform that provides two-way real-time, accurate translations to bridge the communication gap between the hearing impaired and the hearing world**, **fostering inclusivity.**

Motivation

Based on the data we have analyzed from some previous studies conducted we have found that approx. **70-80% of the deaf/mute people who use sign language to speak say that they face significant communication barriers in their daily interactions with normal people**. This leads to **misunderstandings**, **isolation** & **limited participation** in most of the cases.
Thus, we have identified the following customer segments and considered the following hypothesis:

Customer Segments

1. **Mute, Deaf and Hearing-impaired Individuals:**
   Hypotheses:

- Demographics : Mute & deaf people, Age: 13-50 yrs., different Professions
- Behaviours: Use unique methods for communication such as Sign language, lip-reading etc., Use smartphones, PCs etc.
- Attitudes: Strongly value inclusivity and accessibility. Tech-savvy

2. **Hearing individuals interacting with Deaf/hard of hearing:**

   This segment includes family, friends, colleagues, & other individuals interacting frequently with deaf people

   Hypothesis:

   - Demographics: Age and profession vary widely. Families with deaf members, friends, colleagues etc
   - Behaviors: Regularly communicate with deaf individuals, and have lesser familiarity with sign language or lipreading
   - Attitudes: Appreciate smoother communication with their deaf counterparts. Open to learning and adopting new methods.

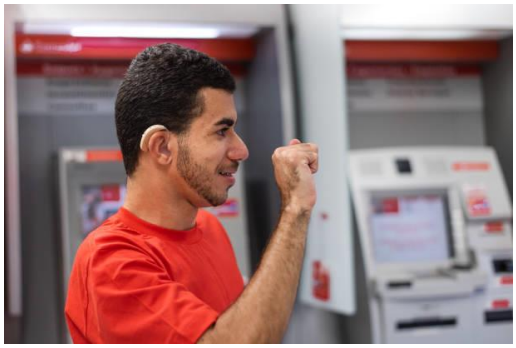
3. **Educational Institutions:**

   Hypothesis:

   - Demographics: Staff and students in schools, colleges, and universities. Age ranges would cover children to adults from 12-55 yrs
   - Behaviors: Actively engage in educational and communication activities involving mute and deaf individuals. Seek inclusive learning environments.
   - Attitudes: Want to enhance the learning experience for mute and deaf students and facilitate effective communication within the educational setting.

4. **Government institutions**:

- Government offices, government banks, Airports etc;
  Hypothesis
  Demographics: Government employees working on service/help desk in different Government offices and backgrounds.
  Behaviours: Regularly interact with a broad range of citizens, including those with speaking & hearing impairments.
  Attitudes Prioritize quick & effective communication with all individuals seeking gov. services.



5. **Corporate Organizations:**

- Businesses that interact with deaf customers, deaf employees.
  Hypothesis:
  - Demographics: Businesses or corporate offices.
  - Behaviors: Regularly interact with deaf colleagues or customers.
  - Attitudes: Value an inclusive workplace and better communication with a diverse customer base. Prioritize accessibility as part of their corporate social responsibility.

Research Design:

Sample Size: Each team member interviewed 5 customers, resulting in a diverse sample size representing different customer segments.

Interview Questions: The team posed the following open-ended & past focused questions to deaf & normal speaking people who interact regularly in each of the target customer segments, focusing on communication preferences, challenges faced, common scenarios of communication barriers, and strategies to address these challenges.

- Q1: How did you communicate last time with a normal speaking person / (Deaf person)? What is your preferred mode of communication with normal-speaking people (Deaf person)?

- Q2: Tell me about a time when you experienced a communication barrier with normal-speaking people (Deaf people). What were the situations/ places where you faced communication barriers? How did you feel about that?

- Q3 How did you overcome/resolve these communication barriers? What do you think will help you bridge the communication gaps?

Interview Setting: Interviews were conducted in different locations such as educational institutions (CSSH, Northeastern University) , community centers such as Mass Association for deaf & hard of hearing, Boston. & Bpston ASL Meetup etc. to capture a range of perspectives.

Interviews with Deaf, Mute, and Hearing-Impaired Individuals:

These interviews provided insights into

- The communication preferences such as American Sign Language (ASL), using language interpreters, lip reading, writing etc.
- Challenges, and experiences of deaf and hearing-impaired individuals during group discussions, doctor appointments, meetings etc.
- They also shed some light on what they do currently to solve their challenges such as basic ASL training, using language interpreters etc.

This helped us in validating the hypothesis that this customer segment faces unique communication barriers as they cannot communicate quickly and efficiently.

Interviews with Hearing Individuals Interacting with Deaf/Hard of Hearing:

- Interviews with hearing individuals interacting with deaf/hard of hearing explained on the challenges faced in understanding of deaf individuals' communication needs.
- These interviews helped validate the hypothesis that hearing individuals interacting with the deaf require support and awareness of diverse communication methods.

Insights from Educational Institutions, Government Institutions, and Corporate Organizations:

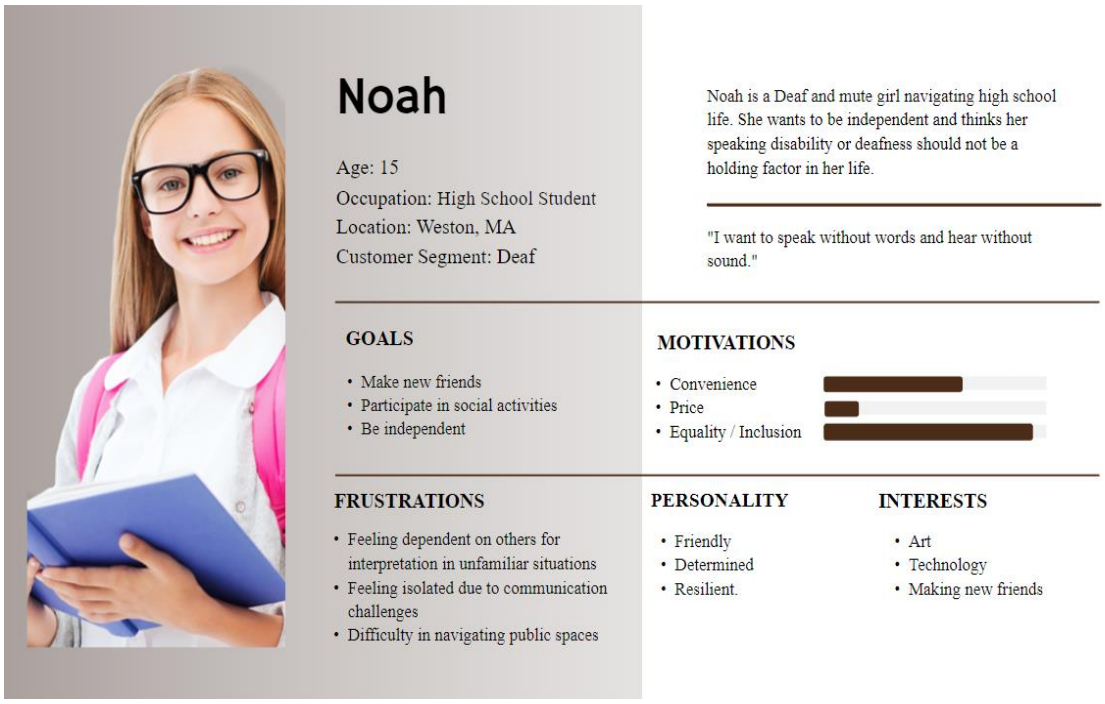
- By interviewing individuals from these sectors, the team gathered evidence on the communication challenges faced in these environments during lectures, meetings, at helpdesks etc. validating the hypothesis that there is a need for improved communication solutions in these settings as quick & effective communication in crucial.

Early Adopters vs. Mainstream Users:

Early Adopters:
Characteristics: Early adopters are individuals among our target segments who are young, tech-savvy and typically more open to trying new technologies, products, or services at an early stage of their development.
Behaviour: They are willing to experiment with innovative solutions to address their needs or challenges in communication.
Communication Preferences: They are likely to embrace new communication technologies and methods that offer improved accessibility and inclusivity for deaf and hearing-impaired individuals.
Role: Early adopters can play a crucial role in providing feedback, testing prototypes, and advocating for the adoption of innovative communication solutions within their communities.

Mainstream Users:
Characteristics: Mainstream users are individuals from our target segments who may be more cautious and require more evidence and validation before adopting new technologies or solutions.
Behaviour: They tend to wait for proven effectiveness, user reviews, and widespread acceptance before embracing a new product or service.
Motivation: Mainstream users prioritize reliability, usability, and practicality in the solutions they adopt, seeking assurance of tangible benefits.
Communication Preferences: Mainstream users may require clear evidence of the effectiveness and usability of communication technologies like SignSense before incorporating them into their regular communication practices.
Role: Mainstream users represent a larger segment of the population and their adoption of new technologies can significantly impact the mainstream acceptance and success of innovative solutions.

User Personas:



Sophia

Age: 16
Occupation: Student
Location: Waltham, MA
Category: Normal Speaking

Sophia is a new friend of a deaf individual (Noah). Sofia doesn't understand sign language. She's eager to improve their communication and build a stronger friendship by learning how to effectively communicate with him.

"Creating a space where everyone feels empowered and unstoppable "

GOALS

- Make good friendships
- Ensure mutual understanding and support

MOTIVATIONS

- Convenience
- Price
- Equality / Inclusion

FRUSTRATIONS

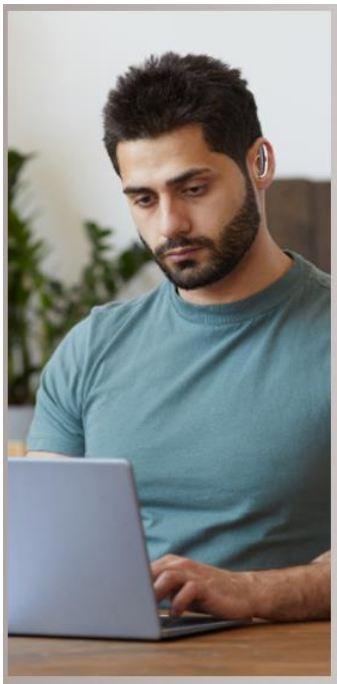
- Feeling helpless.
- Misunderstandings .
- Learning new languages

PERSONALITY

- Impatient
- Helpful
- Compassionate

INTERESTS

- Music & arts
- Technology
- Making new friends



Michael

Age: 27
Occupation: Product Manager
Location: LA, California
Category: Mute/ Deaf Individual

Michael is a Deaf individual who is a product manager at a tech company. He wants to actively participate in corporate projects.

" Inclusion isn't just a goal; it's a necessity for success.

GOALS

- Professional growth and recognition
- Contributing meaningfully to projects
- Communicate ideas clearly and effectively

MOTIVATIONS

- Convenience
- Price
- Equality / Inclusion

FRUSTRATIONS

- Feeling overlooked in team discussions.
- Limited understanding of sign language by colleagues
- Inaccessible public spaces for deaf

PERSONALITY

- Ambitious
- Determined
- Proactive.

INTERESTS

- Technology
- Networking
- Travelling

Marcy

Age: 40
Occupation: Director of Diversity, Equity, and Inclusion (DEI) activities at Texas instruments
Location: Austin, Texas
Category: Normal speaking

Marcy is a dynamic HR professional with a focus on championing diversity and inclusion initiatives at Texas instruments. Passionate about creating an inclusive work culture that celebrates differences and ensures every employee feels valued.

" It's not just about innovation; it's about creating a workplace where every individual's unique contributions are acknowledged and valued.".

GOALS

- Implementing HR policies and practices.
- Facilitating effective communication
- Creating a workplace environment that supports the success of employees from all backgrounds.

MOTIVATIONS

- Convenience
- Price
- Equality / Inclusion

FRUSTRATIONS

- Communication barriers within diverse teams.
- Addressing misunderstandings related to inclusion.

PERSONALITY

- Ambitious
- Determined
- Proactive

INTERESTS

- Technology
- Networking
- Making new friends

Prof. Greg

Age: 52
Occupation: Graduate school professor
Location: Manhattan, New York
Category: Normal Speaking

Prof. Greg is a graduate school professor at NYU who has some deaf students in his classes. He also works as an advisor to Ed-tech research projects at NYU.

"Learning knows no boundaries, and neither should communication."

GOALS

- Create an inclusive learning environment
- facilitate meaningful discussions

MOTIVATIONS

- Convenience
- Price
- Equality / Inclusion

FRUSTRATIONS

- Inconvenience & hurdles while teaching
- Learning new languages

PERSONALITY

- Passionate
- patient
- Innovative

INTERESTS

- Technology
- Research
- Reading

Unmet Needs

Unmet Needs Hypotheses and Evidence Collection:

Mute & Deaf Individuals:

Hypothesis 1: I believe mute & deaf individuals face difficulties in accessing real-time communication support and sign language interpreters in workplace settings, hindering effective collaboration and productivity.

Evidence: Interviews with hearing-impaired individuals in professional spaces revealed the lack of interpreters and accessible technologies as common challenges. This constraint can impact performance and communication effectiveness.

Hypothesis 2: I believe deaf individuals experience challenges in understanding verbal communication in fast-paced environments, leading to feelings of frustration and exclusion.

Evidence: Interviews with deaf individuals highlighted struggles in educational, professional, and social settings due to rapid speech, lack of visual aids, and absence of interpreters . These challenges impact mental well-being and social interactions.

Educators and Students in Educational Institutions:

Hypothesis: I believe educators and students interacting with mute & deaf people in educational institutions encounter barriers in facilitating and participating in inclusive communication practices, affecting learning outcomes and engagement.

Evidence: Interviews with teachers and students highlighted communication challenges during lectures, group projects etc . Lack of awareness and accommodation can lead to feelings of isolation among deaf and hearing-impaired students.

Government Institutions (e.g., Government Offices, Banks, Airports):

Hypothesis: I believe government institutions face challenges in providing effective communication access for deaf individuals, leading to difficulties in accessing services and information.

Evidence: Interviews with deaf individuals accessing government services highlighted the lack of sign language interpreters, limited availability of visual aids, and communication barriers at government offices , banks, airports etc . These challenges can result in delays, misunderstandings, and reduced accessibility for deaf people seeking these services.

Corporate Organizations (Businesses Interacting with Deaf Customers/Employees):

Hypothesis: I believe corporate organizations encounter challenges in effectively communicating with deaf customers and employees, leading to barriers in customer service, team collaboration, and inclusion.

Evidence: Interviews within corporate settings revealed gaps in communication support, limited awareness of deaf communication needs, and the absence of inclusive practices . These challenges can impact customer satisfaction, employee engagement, and overall business performance

Existing Solutions

- There is only one app on google play store Sign language translator AI which only supports Korean language and their sign language to speech translator tool is not working.
- Our major competitor is HandTalk translator, which converts text/audio into sign language but they do not convert sign language into narrative English/other language.
- There is one more paid application ASL Translator which is similar to HandTalk translator.
- Motionsavyy, a deaf-owned company tried to solve a similar problem through a separate tablet device and software supporting it. However the company went bankrupt due to a leadership crisis.

Feature Description	SignSense Empowering inclusive communication **SignSense**	**Hand Talk**	**Sign Language Translator AI**	**ASL Translator**
Price	Starting from $3.99/mo. based on add-on features, No. of licenses & customization	$4.99/mo. 41.99/Yr.	-	$4.99 One-time purchase
Free Tier	Free with ads & limited features	Free with ads & limited features	Free	No! its Paid
Real-time translation from ASL & narrative languages	✓	✗	App supports only Korean Sign language & it's not working	✗
Real-time translation from Text/Speech to ASL	✓	✓	✗	✓
Voice Recognition	✓	✓	✗	✗
Real-time captioning	✓	✓	✗	✗
Offline Mode	✓	Yes (limited features)	✓	✗
Cross-Platform Compatibility (Mobile, Web)	Mobile & Web Application	Only Mobile	✗	✗
Multilingual Support	✓	✗	✗	✗
Learning Resources (ASL)	✓	✓	✗	✓

App integrations (e.g., with video conferencing platforms)	✓	In Beta Test	✗	✗
Customization Options (for translation speed, font size, etc.)	✓	✓	✗	✗
Feedback Mechanism from users to improve translation accuracy.	✓	✓	✗	✗
Translation History	✓	✓	✗	✓
Customizable Avatar options	✓	✓	✗	✗
Privacy & security features	✓	✓	✗	✗

Differentiation

SignSense has several differentiating factors from other available/existing options:

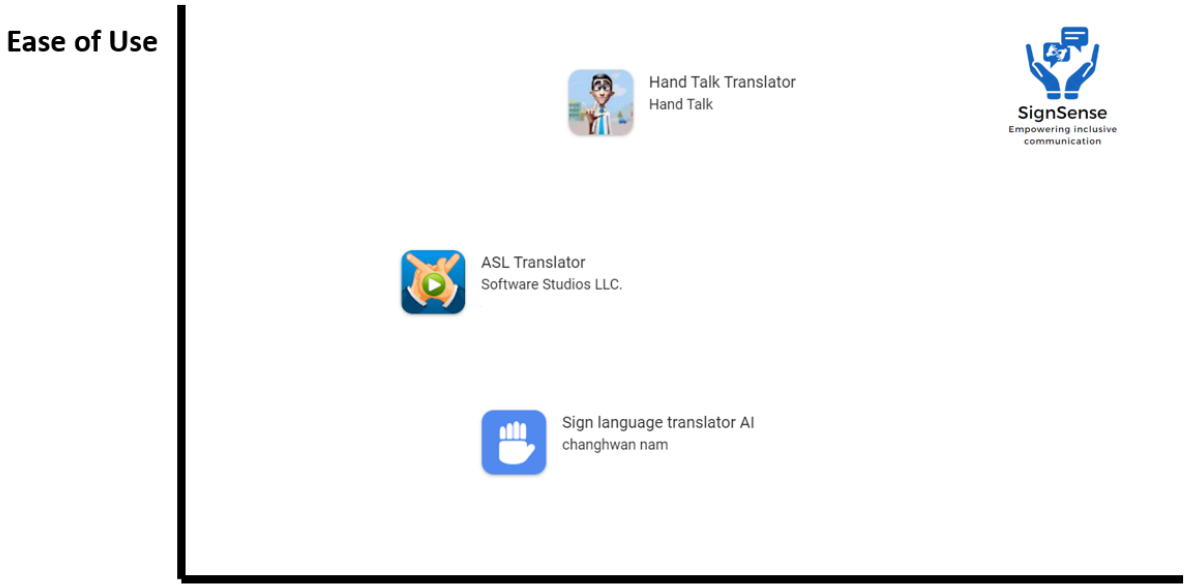
Innovative features:

- SignSense offers real-time translation from ASL to narrative languages which is not there in any of the available solutions.
- Voice recognition and real time captioning is also anavailable in most of them.
- Multilingual support sets SignSense apart from competitors and caters to a broader audience.

User-Centric Approach:
Features such as cross-platform compatibility on mobile & web; product tours, learning resources; customization options for speed, fonts etc; App integrations and customized avatar options emphasize a user-centric approach, enhancing user experience and engagement.

Commitment to Improvement:
By prioritizing the Feedback Mechanism from users to improve translation accuracy and give other feedback for improvement, our team shows a dedication to refining the app based on user needs and preferences.

Focus on Privacy and Security: With robust privacy and security features, SignSense is focusing on instilling trust in users regarding the protection of their data, a critical aspect in today's digital landscape.



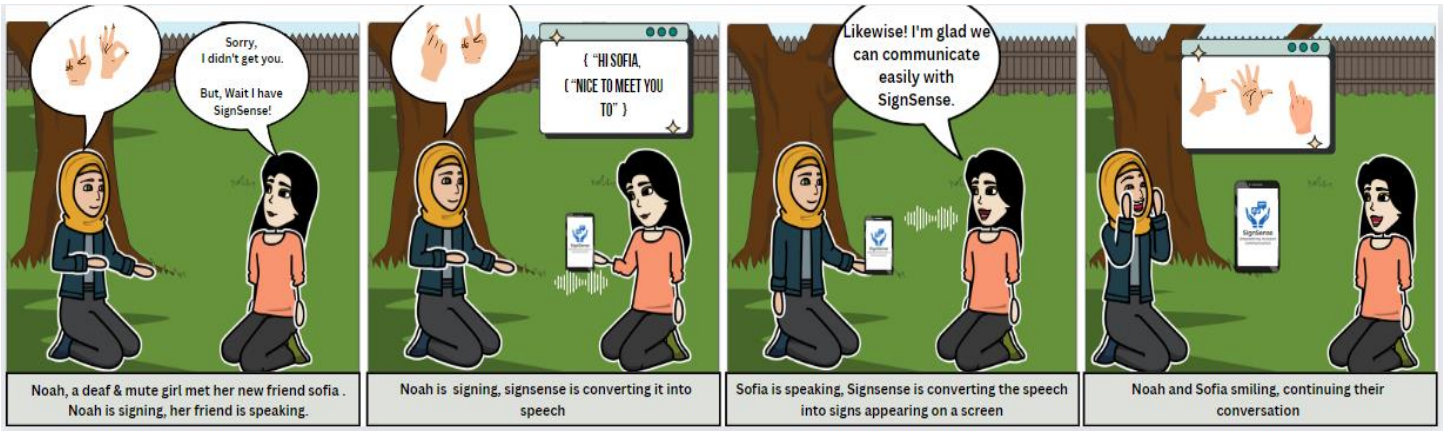
Why Now?

Launching SignSense now is a strategic move due to:

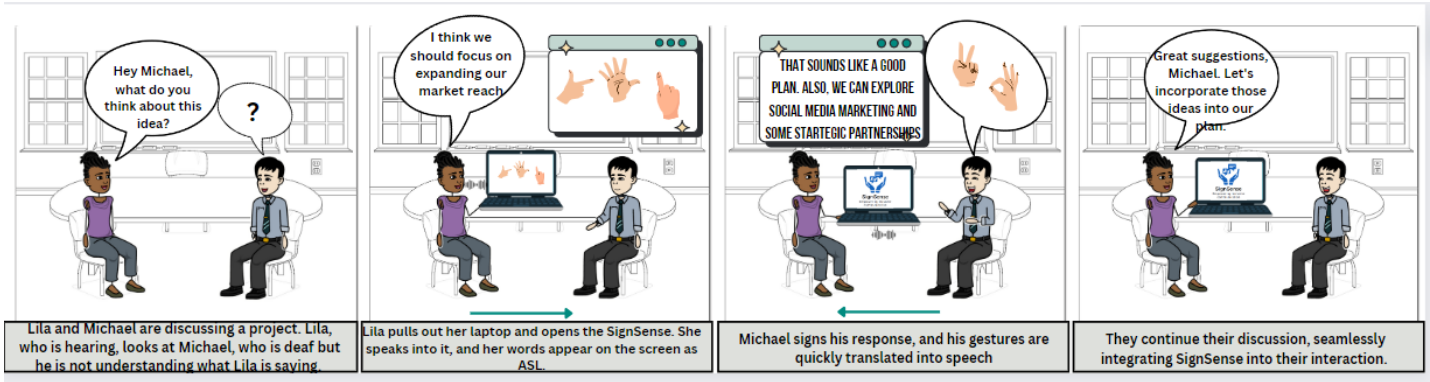
- Technological Advancements & availability: AI and image processing advancements are readily available which enable real-time translation of Sign language.
- Device Accessibility: Smartphones and PCs are now ubiquitous, providing a ready user base for signSense.
- Less Competition: A less crowded market landscape gives SignSense a competitive edge.
- Accessibility & Empowerment: Meeting the growing societal focus on creating more inclusive world for the deaf community fostering accessibility and empowerment.
- Market Potential: Significant market demand and growth potential.
- Partnership Opportunities: Potential collaborations organizations focused on accessibility and inclusivity for wider impact and reach.

Use Cases

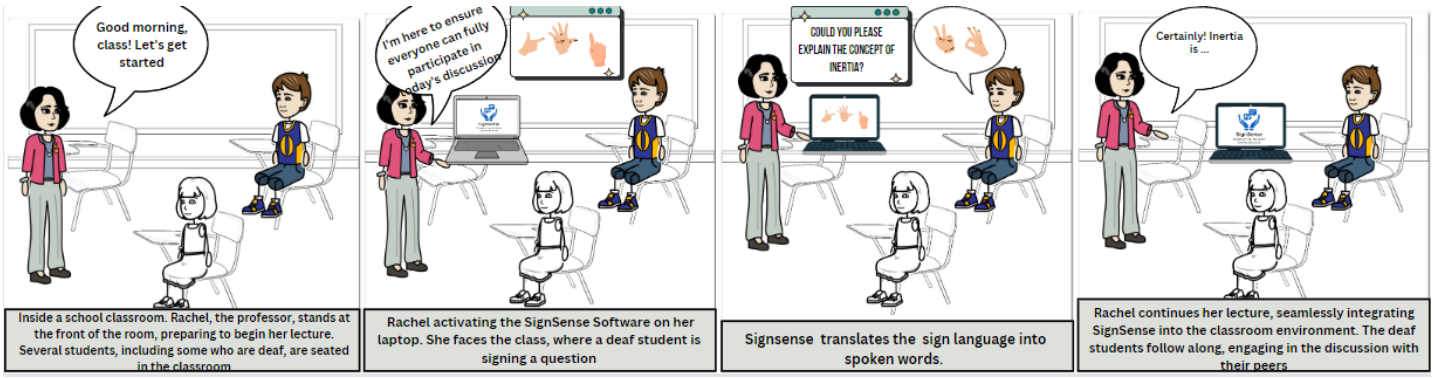
1. Noah, a deaf & mute girl met her new friend Sofia. Noah is signing, her friend is speaking. Noah is signing, signsense is converting it into speech. Sofia is speaking, Signsense is converting the speech into signs appearing on a screen. Noah and Sofia smiling, continuing their conversation.



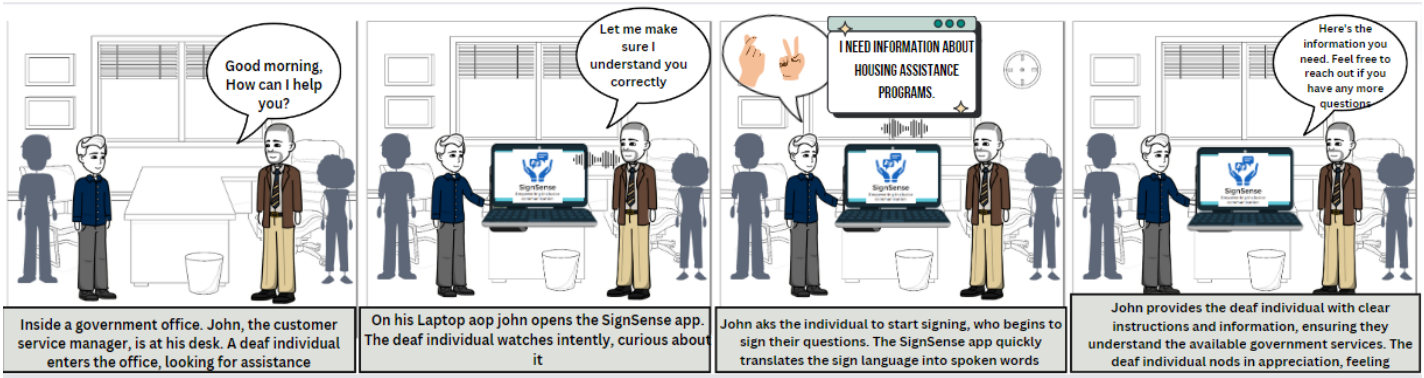
2. Lila and Michael are discussing a project. Lila, hearing, looks at Michael, who is deaf, but does not understand what Lila is saying. Lila pulls out her laptop and opens the SignSense. She speaks into it, and her words appear on the screen as ASL. Michael signs his response, and his gestures are quickly translated into speech. They continue their discussion, seamlessly integrating SignSense into their interaction.



3. Inside a school classroom. Rachel, the professor, stands at the front of the room, preparing to begin her lecture. Several students, including some who are deaf, are seated in the classroom. Rachel activates the SignSense Software on her laptop. She faces the class where a deaf student is signing a question. Signsense translates sign language into spoken words. Rachel continues her lecture, seamlessly integrating SignSense into the classroom environment. The deaf students follow along, engaging in discussion with their peers.



4. Inside a government office. John, the customer service manager, is seated at his desk. A deaf individual enters the office, looking for assistance. Close-up of John pulling out his Tablet and opening the SignSense app. The deaf individual watches intently, curious about the device. John holds up facing the deaf individual, who begins to sign their questions. The SignSense app quickly translates the sign language into spoken words displayed on the screen. John provides the deaf individual with clear instructions and information, ensuring they understand the available government services. The deaf individual nods in appreciation, feeling empowered and supported.



Market Size

- Individuals who are disabling deaf or hard of hearing are around 11 million in USA
- Out of these, deaf or hard of hearing individuals who use ASL as their primary means of communication are about 2 million in USA.
- In a typical day, a deaf person interacts with 10-12 people.
- Educational institutions, Banks, Healthcare organizations, Government bodies, Businesses that serve deaf or hard-of-hearing individuals require ASL interpretation services. That is around 150,000

Hence the total addressable market can be estimated to be approximately 20 million users (about the population of Texas).

Caveats / Risks / Key Dependencies

Risk	Description	Mitigation Plans
Privacy Concerns	**1. Data Collection and Storage**: The app will likely collect user data such as hand movements, facial expressions, and potentially audio for translation. This data can be highly sensitive and needs proper handling. **2. Misuse of Translated Data**: Translated content, both spoken and Visual could be misused if intercepted or leaked for discrimination, profiling, or even identity theft. **3. Control / Trust issues over data:** Users might feel they have limited control over the data collected about them.	1. Implement strong security measures like encryption for data storage and transmission. 2. Clearly outline data collection practices and obtain explicit user consent. 3. Anonymize data whenever possible and only use it for specific translation purposes. 4. Ensure compliance with relevant data privacy regulations like GDPR and CCPA. 5. Develop robust access controls and limit data access only to authorized personnel.
Legal/Patent Risks	**1. Infringement of Existing Patents**: Some aspects of the translation technology or app functionality might infringe on existing patents owned by others. **2. Accessibility Compliance Issues:** Failing to comply with accessibility regulations like the Americans with Disabilities Act (ADA) could lead to legal challenges.	1. Conduct thorough patent searches before development to avoid infringement. 2. Seek legal counsel/ experts to ensure compliance with relevant laws and regulations. 3. Design the app with accessibility in mind, incorporating features for diverse users, such as text-to-speech and voice control options.
Piracy Concerns	The app's functionality, code or user interface could be illegally copied and distributed, impacting revenue and user trust.	1. Implement robust anti-piracy measures like digital rights management (DRM) and code obfuscation. 2. Use secure licensing mechanisms and actively monitor for unauthorized distribution.
Interdependencies with Internal Systems	**1. Functionality issues** due to integration challenges in backend and frontend systems. **2. operational challenges** due to incompatible data formats. **3. System downtime**: Disruptions or	1. Early planning and collaboration: Involve relevant teams like IT and development early in the planning process to identify potential integration challenges. 2. Standardization and API development: Consider standardizing data formats and

	maintenance of internal systems could affect SignSense functionality.	developing well-defined APIs to facilitate seamless communication 3. Thorough testing: Conduct comprehensive testing of the integration process in various scenarios to ensure smooth functionality and identify any compatibility issues before launch. 4. Redundancy and failover plan: Implement backup systems and failover mechanisms to minimize downtime
Availability of Third-Party Complements	**1. Reliance on External APIs:** The app might rely on external APIs for specific functionalities like image processing, Motion sensing, speech recognition or text-to-speech synthesis. These APIs could become unavailable or change in the future, impacting the app's functionality. **2. Integration with 3rd party Software** such as MS Teams, Zoom etc. **3. Security Risks associated with Third-Party Services**: Utilizing third-party services introduces additional security risks, as the app becomes dependent on their security practices.	1. Diversify and have backup options for crucial third-party services to minimize dependence on a single provider. 2. Evaluate the security and reliability of third-party services before integration. 3. Implement robust security protocols and data encryption on both ends of the integration point.

Strategic Considerations

The proposed solution, SignSense, aligns seamlessly with our organization's vision to empower communication and enhance accessibility. SignSense directly supports our overarching goal by breaking down communication barriers, particularly for individuals with hearing impairments. By providing real-time sign language translation and fostering inclusive communication, SignSense contributes directly to our commitment to making communication more accessible. It is in perfect harmony with our organization's strategy of leveraging technology to create solutions that empower individuals and promote inclusivity in various aspects of their lives.

Go/No Go Recommendation

We should go ahead with SignSense because of the following reasons:

1. Addressing a Real Need: We have **validated our target audience's needs** through research and confirmed the **existence of a real problem**, which is the communication barrier faced by the deaf and hard-of-hearing community.

2. Technological Readiness: The **infrastructure** required to support SignSense, including advancements in **AI (Machine Learning and Image Processing)**, is now **readily available** and robust enough to deliver accurate translations.

3. Competitive Advantage: The **absence of significant competition** in the sign language translation application market presents a **unique opportunity** for SignSense to establish itself as a **leader in this emerging field**.

4. Market Potential: The market for sign language translation applications is expected to experience **significant growth** in the coming years, presenting a sizeable **business opportunity** for SignSense.

5. Societal Impact: SignSense aligns perfectly with the growing **societal focus on inclusivity and accessibility**. By launching this application, we can **empower the deaf and hard-of-hearing community** and contribute to building a more inclusive world for everyone.

References

- How many deaf people live in the United States? - National Deaf Center
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